



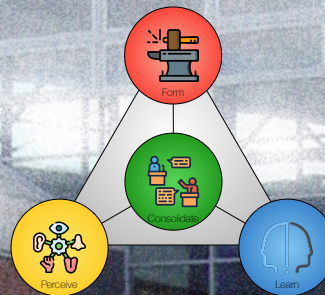
AALBORG UNIVERSITY
DENMARK

8 INNER ENVIRONMENT

ESSENCE – CHAPTER 12

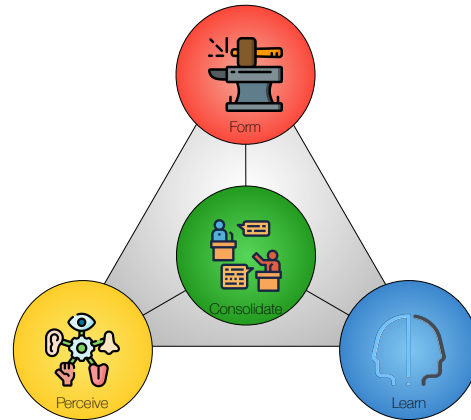
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Agenda

- Exercise 7 Follow-up.
- Chapter 10: Inner Environment.
 - Inner Environment.
 - Near-decomposability as a general strategy.
 - VUCA Strategies for Inner Environment.
- Next time:
 - Exercise 8: Inner Environment.
 - Lecture 9: Evolvability.

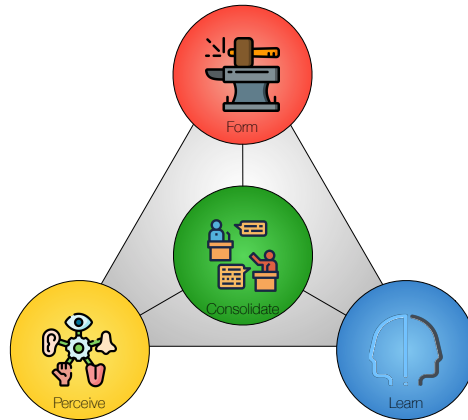


Follow-up: Exercise 7

Exercise 7 (Strategy & Outer Environment)

This exercise is about VUCA strategies for the Outer Environment.

- Discuss VUCA challenges for your project (see Table 10.1). Which of these challenges are most likely to affect your project?
- Discuss VUCA strategies for your Outer Environment (see Table 11.2). Are they useful? How? Suggest alternative strategies if they are not useful.
- How could you use these strategies actively in your project?



Strategy

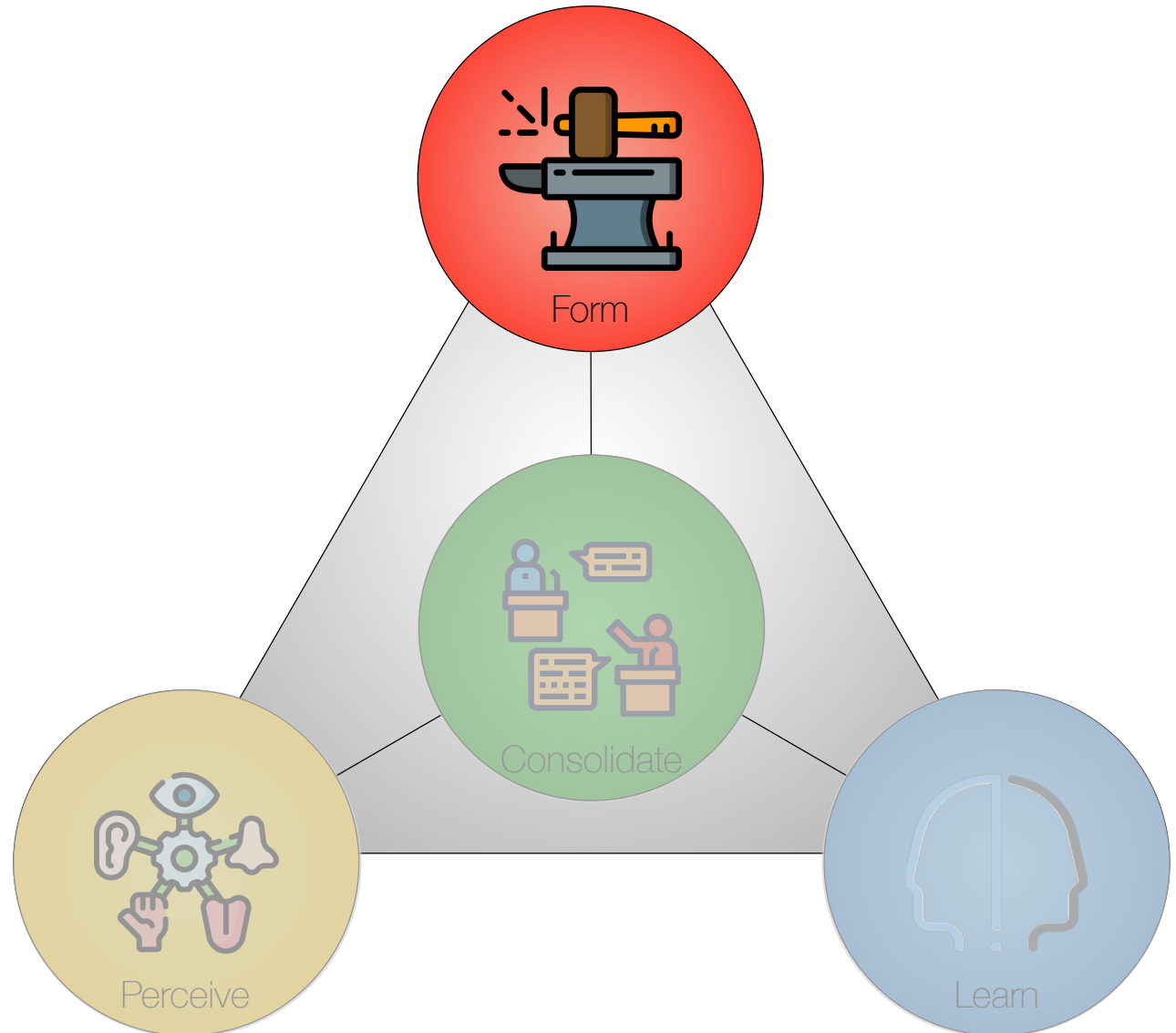


In preparing for battle, I have always found that plans are useless, but planning is indispensable.

Dwight D. Eisenhower (1890–1969)

Four Core Activities

- *Perceive* – understanding the problem and its context.
- *Form* – designing the constructive parts of a solution.
- *Consolidate* – integrating the design contributions into a clear whole that addresses the situation.
- *Learn* – appraising the design.



Three Levels of Abstraction



Rationale – the logical basis that makes actions meaningful and helps reason about strategy and tactics for solving the overall problem.

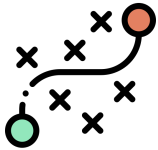


Strategy – the **overall plan** for addressing the problem, including its **scope** and **context**, the essential parts needed to build the solution, and any limitations or constraints that might influence its feasibility. At the strategy level, we also evaluate the potential for future **growth**.



Tactics – actions planned in accordance with the strategy to achieve specific ends.

Contribution Terminology

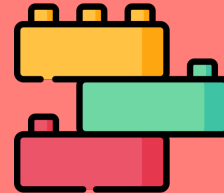


Capabilities



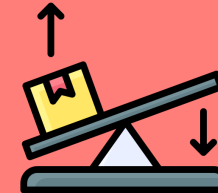
Capabilities are the features that are designed to address manifestations.

Inner Environment



The inner environment is what provides the capabilities.

Leverage



Leverage points allow for the most valuable capabilities.

Changing Conditions

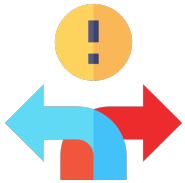
Essence is designed for conditions in which the inner or outer environment will change **significantly** as we learn more or as **external factors** affect our project.

- Learning more could provide **new insights** about our customers, users, or ongoing projects that **rely** on us or on which we **rely** in our project.
- We cannot assume that the situation will remain constant throughout the project.
- Therefore, we cannot rely on a **master plan** that will remain viable throughout the project.

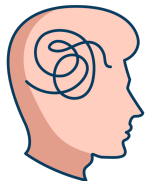
VUCA



- *Volatility* – i.e., conditions that change rapidly or unpredictably.



- *Uncertainty* – i.e., needs and solutions that are unsettled.



- *Complexity* – i.e., external or internal parts that form intricate meshes.



- *Ambiguity* – i.e., causes and effects that are mixed up.

Strategic Challenges for Design

		KNOWLEDGE ABOUT THE SITUATION	
ABILITY TO PREDICT THE RESULTS OF ACTIONS	HIGH	COMPLEXITY  External or internal parts that form intricate meshes. The situation or contribution has many interconnected parts and variables. Some information may be available, but the volume can be overwhelming.	VOLATILITY  Conditions that change rapidly or unpredictably. The situation or contribution is unstable or may change unexpectedly, but is not necessarily hard to understand. Knowledge about the change is often available.
		AMBIGUITY  Causes and effects that are mixed up. Causal relationships are completely unclear. The logic of the situation changes with our actions, and stakeholders might be unsure of their preferences.	UNCERTAINTY  Needs and solutions that are unsettled. Basic causes and effects are known, but there is a lack of knowledge about possible changes. The future is, therefore, unpredictable despite available information.
	LOW		
		Low	HIGH

Inspired by (Bennett and Lemoine, 2014)

The Inner Environment

The inner environment is our contribution – the substance and organization of our design and implementation.

This definition is based on Simon's definition: *The substance and organization of the artifact itself* (Simon, 1996, p. 6).

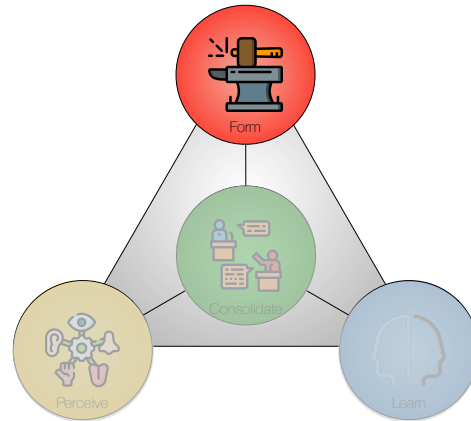
- The word artifact is Simon's term for what we call *design*.
- The artifact is the *meeting point* between the outer and inner Environment.

In Essence, the inner environment is the *architecture* and the *primary components* used to exploit the *leverage* and implement the desired *capabilities*.

Mutual Adaptation

The relationship between the inner and outer environment is a matter of *mutual adaptation*.

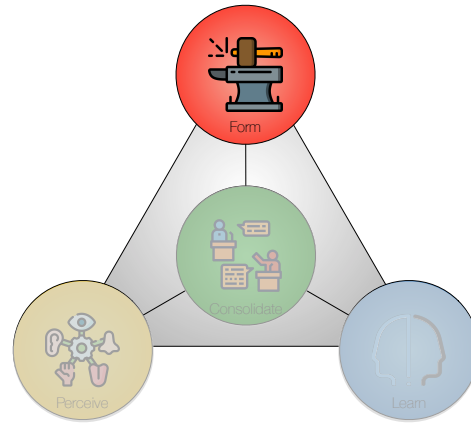
Such adaptation is inherently *dynamic* and also *iterative* as we learn more about both environments through *transactions*.



Near-decomposability as a general strategy

Evolution of Complex Systems

- There once were two watchmakers, named **Hora** and **Tempus**, who manufactured very fine watches. Both of them were highly regarded, and the phones in their workshops rang frequently—new customers were constantly calling them. However, **Hora** prospered, while **Tempus** became poorer and poorer and finally lost his shop. What was the reason?
- The watches the men made consisted of about 1,000 parts each. **Tempus** had so constructed his that if he had one partly assembled and had to put it down—to answer the phone, say—it immediately fell to pieces and had to be reassembled from the elements. The better the customers liked his watches, the more they phoned him and the more difficult it became for him to find enough uninterrupted time to finish a watch.
- The watches that **Hora** made were no less complex than those of **Tempus**. But he had designed them so that he could put together subassemblies of about ten elements each. Ten of these subassemblies, again, could be put together into a larger subassembly; and a system of ten of the latter subassemblies constituted the whole watch. Hence, when **Hora** had to put down a partly assembled watch to answer the phone, he lost only a small part of his work, and he assembled his watches in only a fraction of the man-hours it took **Tempus**.



VUCA

Inner Environment

VUCA & Leverage Points

CATEGORIES	CHARACTERISTICS
<i>Technology</i>	Scientific knowledge, machinery, and equipment on which a design might be based.
<i>Components</i>	Rare equipment, sensors, or actuators used actively in building a solution and accessed via crafted interfaces.
<i>Information</i>	Used for collecting, storing, disseminating, and/or aggregating information to be accessed via crafted interfaces.
<i>Human resources</i>	Specific users, categories of users, experts, authorities, social or professional networks, organizations, or others with particular roles or functions designed to build a solution. Cyberhuman systems are prominent examples of systems where internal people are leverage points.

Four leverage categories (chapter 5).

Leverage points might be affected by VUCA
—this could reduce (or increase) survivability

VUCA in the Outer Environment is Linked to the Inner Environment

CATEGORIES	CHARACTERISTICS
<i>External services</i>	Services that are used without modification and reduce what is left for us to deal with.
<i>External implements</i>	Equipment, sensors, or actuators available in the outer environment and accessed via trivial interfaces.
<i>External repositories</i>	Repositories for collecting, storing, disseminating, and/or aggregating information via trivial interfaces.
<i>External people</i>	Generally available people, professionals, stakeholders, authorities, networks, organizations, etc.

Four outer environment categories (chapter 4).

The outer environment might be affected by VUCA—
this could reduce (or increase) survivability

VUCA Strategies for Inner Environment

		KNOWLEDGE ABOUT THE SITUATION	
		LOW	HIGH
ABILITY TO PREDICT THE RESULTS OF ACTIONS	HIGH	COMPLEXITY  External or internal parts that form intricate meshes. STRATEGIES • Modularize • Divide & conquer	VOLATILITY  Conditions that change rapidly or unpredictably. STRATEGIES • Build generic capabilities • Use microservice architecture
	LOW	AMBIGUITY  Causes and effects that are mixed up. STRATEGIES • Experimentation (MVP) • Problem-solving as natural selection	UNCERTAINTY  Needs and solutions that are unsettled. STRATEGIES • Information hiding (Parnas) • Isolate uncertainties in modules

V: *Generic* means reusable, lego parts. Make sure parts can contribute to other purposes

U: *Information hiding*. Make sure critical parts can be changed without side effects

C: *Modularize*: Simplify through abstraction. *Divide & conquer*: Pivot, zoom-in, simplify through reduction

A: *MVP*: try things out, select promising contributions. Search for capabilities that are valued. Test



Volatility

Volatility refers to conditions that change rapidly or unpredictably.

- *Outer environment:* Sudden disruptive changes in legislation, technological breakthroughs, market conditions, revenues, the economy, and competition, among other things.
- *Inner environment:* Sudden disruptive changes in leverage points, such as the technologies, components, information, or people available for building contributions.



Volatility

Designing for conditions that change rapidly or unpredictably.

Strategies:

- Build *generic* capabilities.
 - Common or nonspecific capabilities that are easily adapted to changing conditions.
 - Design components with lines of tearing that allow them to endure in changing environments (inspired by Augier & Sarasvathy p187)
- Use *microservice* architecture.
 - A collection of *loosely coupled*, fine-grained services.
 - Services that can be developed and deployed *independently* of others.



Uncertainty

Uncertainty refers to unsettled needs and solutions.

- *Outer environment:* Unsettled disruptive changes in legislation, technological breakthroughs, market conditions, revenues, the economy, and competition, among other things.
- *Inner environment:* Unsettled disruptive changes in leverage points, such as the technologies, components, information, or people available for building contributions.



Uncertainty

Designing for needs and solutions that are unsettled.

Strategies:

- Use *Information hiding* (Parnas)
 - Hide difficult design decisions or design decisions that are likely to change in separate modules.
- *Isolate uncertainties* in modules.
 - Minimize the effects of uncertainty to as few modules as possible.



Complexity

Complexity occurs when external or internal parts of a system form intricate meshes. Complex systems are made up of a large number of parts that interact in a nonsimple way (Simon, 1962, p. 468).

- *Outer environment:* The situation involves many elements (objects or events) that interact in non-simple ways and where the properties of the whole can not easily be inferred.
- *Inner environment:* The contribution involves many elements (objects or events) that interact in non-simple ways and where the properties of the whole can not easily be inferred.



Complexity

Designing when external or internal parts form intricate meshes.

Strategies:

- *Modularize into nearly decomposable systems* (Simon 1962)
- Systems in which the interactions among the subsystems are weak but not negligible.
- Use *divide-and-conquer* patterns.
- Try to split complex tasks into smaller sub-tasks and address them individually.



Ambiguity

Ambiguity describes challenges in which causes and effects are mixed up. Decisions or predictions are difficult to make when actions change the logic of the system in which they occur.

- *Outer environment:* The logic, needs, and constraints of the outer environment are unknown and unavailable, or causal relationships are completely unclear.
- *Inner environment:* The contribution changes the logic, needs, and constraints of the outer environment or leads to unforeseeable changes through development and implementation.

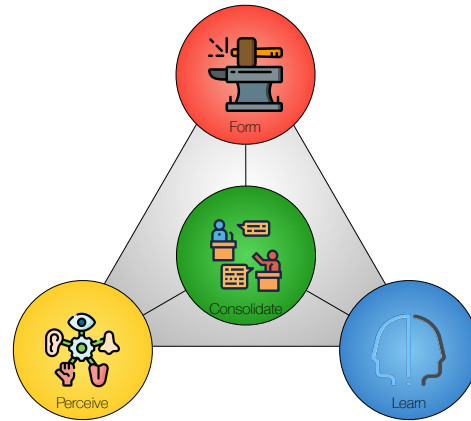


Ambiguity

Designing when causes and effects are mixed up.

Strategies:

- Use *experimentation* (MVP)
 - Use prototypes and similar to maximize user feedback.
- Use problem-solving as a *natural selection* process.
 - *Selective* trial and error.
 - Components that prove *stable* provide new building blocks for further construction (inspired by Simon 1962, p473).
 - These *stable configurations* guide the process of evolution and provide the selectivity essential for *rapid adaptation* (inspired by Simon 1962 p473).



Next Time

Next Exercise and Lecture

- Exercise 8:
 - Inner Environment.
- Lecture 9:
 - Chapter 13 in Essence: Evolvability.

Exercise 8 (Inner Environment)

This exercise is about **near-decomposability** and **VUCA** strategies for the **Inner Environment**.

- The introduction part of chapter 12 (pages 140-141) discusses the use of **conceptual models** for developing and revising design ideas. Which conceptual models do you use to design the architecture and outline primary components in your project?
- Are any of these models **helpful** for modeling near-decomposable designs, as discussed in Section 12.2?
- Discuss VUCA **strategies** for your Inner Environment (see Table 12.1). Are any of them relevant to your project? How? Suggest **alternative** strategies if they are not useful.
- How could you use these strategies **actively** in your project?